

## **SALTAMOL 100 mcg/dose HFA METERED DOSE INHALER**

‘Salbutamol Sulphate as Salbutamol 100mcg/puff’

### **Product Description**

An easily actuated aerosol having plastic actuator of blue color body and light blue color cap embossed “SQUARE” logo on cap. The canister contains a suspension of Salbutamol Sulphate along with a propellant filled in a metallic can which is fitted with a metering valve; imprinted with Batch no., Mfg date and Exp date.

### **Pharmacodynamics**

Salbutamol is a selective  $\beta_2$ -adrenoceptor agonist. At therapeutic doses it acts on the  $\beta_2$ -adrenoceptors of bronchial muscle providing short acting (4-6 hour) bronchodilation with a fast onset (within 5 minutes) in reversible airways obstruction.

#### Special Patient Populations

##### *Children < 4 years of age*

Patients < 4 years with bronchospasm associated with reversible obstructive airways disease, show that salbutamol Evohaler has a safety profile comparable to that in children  $\geq$  4 years, adolescents and adults.

### **Pharmacokinetics**

Salbutamol administered intravenously has a half-life of 4 to 6 hours and is cleared partly renally and partly by metabolism to the inactive 4'-O-sulfate (phenolic sulfate) which is also excreted primarily in the urine. The faeces are a minor route of excretion.

After administration by the inhaled route between 10 and 20% of the dose reaches the lower airways. The remainder is retained in the delivery system or is deposited in the oropharynx from where it is swallowed. The fraction deposited in the airways is absorbed into the pulmonary tissues and circulation, but is not metabolised by the lung. On reaching the systemic circulation it becomes accessible to hepatic metabolism and is excreted, primarily in the urine, as unchanged drug and as the phenolic sulfate.

The swallowed portion of an inhaled dose is absorbed from the gastrointestinal tract and undergoes considerable first-pass metabolism to the phenolic sulfate. Both unchanged drug and conjugate are excreted primarily in the urine. Most of a dose of salbutamol given intravenously, orally or by inhalation is excreted within 72 hours. Salbutamol is bound to plasma proteins to the extent of 10%.

### **Indication**

Salbutamol provides short-acting (4 to 6 hour) bronchodilation with fast onset (within 5 minutes) in reversible airway obstruction.

It is particularly suitable for the relief and prevention of asthma symptoms. Salbutamol should be used to relieve symptoms when they occur, and to prevent them in those circumstances recognised by the patient to precipitate an asthma attack (e.g. before exercise or unavoidable allergen exposure).

Salbutamol is particularly valuable as relief medication in mild, moderate or severe asthma, provided that reliance on it does not delay the introduction and use of regular inhaled corticosteroid therapy.

### **Recommended Dose**

SALTAMOL has a duration of action of four to six hours in most patients.

Increasing use of beta-2 agonists may be a sign of worsening asthma. Under these conditions a reassessment of the patient's therapy plan may be required and concomitant glucocorticosteroid therapy should be considered.

As there may be adverse effects associated with excessive dosing, the dosage or frequency of administration should only be increased on medical advice.

SALTAMOL is administered by the oral inhaled route only.

In patients who find co-ordination of a pressurised metered-dose inhaler difficult a spacer may be used with SALTAMOL.

Babies and young children using the SALTAMOL may benefit from the use of a paediatric spacer device with a face mask.

#### RELIEF OF ACUTE BRONCHOSPASM

##### Adults

100 or 200 micrograms.

##### Children

100 micrograms. The dose may be increased to 200 micrograms if required.

#### PREVENTION OF ALLERGEN OR EXERCISE-INDUCED BRONCHOSPASM

##### Adults

200 micrograms before challenge or exertion.

##### Children

100 micrograms before challenge or exertion. The dose may be increased to 200 micrograms if required.

#### CHRONIC THERAPY

##### Adults

Up to 200 micrograms four times daily.

##### Children

Up to 200 micrograms four times daily.

On demand use of SALTAMOL should not exceed four times daily. Reliance on such supplementary use or a sudden increase in dose indicates deteriorating asthma (see Warnings and Precautions).

**Route of Administration:** Respiratory (Inhalation)

#### Contraindications

Hypersensitivity to the active substance or any of the excipients.

Non-IV formulations of salbutamol must not be used to arrest uncomplicated premature labour or threatened abortion.

#### Warning and Precautions

Patients inhaler technique should be checked to make sure that aerosol actuation is synchronised with inspiration of breath for optimum delivery of drug to the lungs. Patients should be warned that they may experience a different taste upon inhalation compared to their previous inhaler.

Bronchodilators should not be the only or main treatment in patients with severe or unstable asthma. Severe asthma requires regular medical assessment, including lung-function testing, as patients are at risk of severe attacks and even death. Physicians should consider using the maximum recommended dose of inhaled corticosteroid and/or oral corticosteroid therapy in these patients.

The dosage or frequency of administration should only be increased on medical advice. If a previously effective dose of inhaled salbutamol fails to give relief lasting at least three hours, the patient should be advised to seek medical advice.

Increasing use of bronchodilators, in particular short-acting inhaled  $\beta_2$ -agonists, to relieve symptoms, indicates deterioration of asthma control. The patient should be instructed to seek medical advice if short-acting relief bronchodilator treatment becomes less effective, or more inhalations than usual are required. In this situation the patient should be assessed and consideration given to the need for

increased anti-inflammatory therapy (e.g. higher doses of inhaled corticosteroid or a course of oral corticosteroid).

Severe exacerbations of asthma must be treated in the normal way.

Cardiovascular effects may be seen with sympathomimetic drugs, including salbutamol. Patients with underlying severe heart disease (e.g. ischaemic heart disease, arrhythmia or severe heart failure) who are receiving salbutamol should be warned to seek medical advice if they experience chest pain or other symptoms of worsening heart disease. Attention should be paid to assessment of symptoms such as dyspnoea and chest pain, as they may be of either respiratory or cardiac origin.

Salbutamol should be administered cautiously to patients with thyrotoxicosis.

Potentially serious hypokalaemia may result from  $\beta_2$ -agonist therapy, mainly from parenteral and nebulised administration. Particular caution is advised in acute severe asthma as this effect may be potentiated by hypoxia and by concomitant treatment with xanthine derivatives, steroids and diuretics. Serum potassium levels should be monitored in such situations.

As with other inhalation therapy, paradoxical bronchospasm may occur with an immediate increase in wheezing after dosing. This should be treated immediately with an alternative presentation or a different fast-acting inhaled bronchodilator. SALTAMOL should be discontinued immediately, the patient assessed, and if necessary, a different fast-acting bronchodilator instituted for on-going use.

### Interactions with Other Medicaments

Saltamol and non-selective  $\beta$ -blocking drugs such as propranolol, should not usually be prescribed together.

### Pregnancy and Lactation

#### Pregnancy

Safety in pregnant women has not been established. SALTAMOL should not be used during pregnancy unless clearly necessary.

#### Breast-feeding

As salbutamol is probably secreted in breast milk, its use in nursing mothers requires careful consideration. It is not known whether salbutamol has a harmful effect on the neonate, and so its use should be restricted to situations where it is felt that the expected benefit to the mother is likely to outweigh any potential risk to the neonate.

### Side Effects

Adverse events are listed below by system organ class and frequency.

| Organ system                       | Adverse reactions                                                                                                                                                                 | Frequency                                    |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Immune system disorders            | Hypersensitivity reactions including angioedema, collapse, bronchospasm, hypotension, urticaria                                                                                   | Very rare                                    |
| Metabolism and nutrition disorders | Hypokalaemia<br>Potentially serious hypokalaemia may result from $\beta_2$ agonist therapy.                                                                                       | Rare                                         |
| Nervous system disorders           | Tremor, headache<br>Hyperactivity                                                                                                                                                 | Common<br>Very rare                          |
| Cardiac disorders                  | Tachycardia<br>Palpitations<br>Cardiac arrhythmia (e.g. atrial fibrillation, supraventricular tachycardia and extrasystoles)<br>Myocardial ischaemia (see Warning and precaution) | Common<br>Uncommon<br>Very rare<br>Not known |
| Vascular disorders                 | Peripheral vasodilatation                                                                                                                                                         | Rare                                         |

|                                                 |                              |           |
|-------------------------------------------------|------------------------------|-----------|
| Respiratory, thoracic and mediastinal disorders | Paradoxical bronchospasm     | Very rare |
| Gastrointestinal disorders                      | Mouth and throat irritation. | Uncommon  |
| Musculoskeletal and connective tissue disorders | Muscle cramps                | Uncommon  |

### **Symptoms and Treatment of Overdose**

The most common signs and symptoms of overdose with SALTAMOL are transient beta agonist pharmacologically mediated events, including tachycardia, tremor, hyperactivity and metabolic effects including hypokalaemia.

Hypokalaemia may occur following overdose with SALTAMOL. Serum potassium levels should be monitored. Lactic acidosis has been reported in association with high therapeutic doses as well as overdoses of short-acting beta-agonist therapy, therefore monitoring for elevated serum lactate and consequent metabolic acidosis (particularly if there is persistence or worsening of tachypnea despite resolution of other signs of bronchospasm such as wheezing) may be indicated in the setting of overdose.

### **Storage Condition**

Store below 30°C. Protect from direct sunlight or heat. Protect from frost. Keep away from contact of eyes.

### **Dosage forms and packaging available**

200puffs/canister

### **Manufactured by:**

SQUARE PHARMACEUTICALS LTD.  
Dhaka Unit, Kaliakoir, 1750 Gazipur, Bangladesh.

### **Product Registration Holder**

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